Chapter 1

six

Remark 1.1 (Physics notation: canonical dimensions and the Weyl rescaling)**.** The Weyl weight of a slot is the canonical mass dimension of its components in the physics notation key after Section 1.2:

$$[A_m]=[A_{ij}]=1,\quad [\lambda_\alpha{}^i]=[\bar\lambda_{\dot\alpha i}]=\tfrac{3}{2},\quad [c]=0,\qquad [A^*_m]=[A^*_{ij}]=3,\quad [\lambda^{*\alpha}{}_i]=[\bar\lambda^{*\dot\alpha i}]=\tfrac{5}{2},\quad [c^*]=4.$$

σ_s is the finite dilatation $x\mapsto sx$ acting on fields of these dimensions, for example

$$(\sigma_s A_m)(x)=s^{-1}A_m(s^{-1}x),\qquad (\sigma_s\lambda_\alpha{}^i)(x)=s^{-3/2}\lambda_\alpha{}^i(s^{-1}x),\qquad (\sigma_s\bar\lambda^{*\dot\alpha i})(x)=s^{-5/2}\bar\lambda^{*\dot\alpha i}(s^{-1}x).$$

With $\omega=\omega(x)\,d^4x$, a local operator is $O=\int d^4x\,\omega(x)f(x)$, the smearing of the composite operator $f(x)$ (a polynomial in the fields and their derivatives at x) of canonical dimension w_O; $E=x^m\partial_m$, and $(E+w_O)O=\int d^4x\,\omega(x)\,(x^m\partial_m+w_O)f(x)$ is the classical scale transformation of f.

Remark 1.2 (Physics notation: the proper-time cutoff under scaling)**.** K_t is the heat kernel of $H=-\partial^2$, ε is the Schwinger proper-time cutoff, of dimension length^2, and $\varepsilon^{-1/2}$ the corresponding momentum cutoff: every component of $P(\varepsilon,\infty)$ carries $\int_\varepsilon^\infty e^{-t|p|^2}dt=e^{-\varepsilon|p|^2}/|p|^2$ (Proposition **??**). Proposition **??** states that the rescaling $x\mapsto sx$ of lengths is the rescaling $\varepsilon\mapsto s^2\varepsilon$ of the cutoff. In Lemma **??** the coefficients of ε^{-1} are power (quadratic) divergences, multiplied by s^{-2}, the one of $I^{CT}[\varepsilon]$ being a multiple of the mass-type term $\int\big[(A_m,A_m)+\frac14(A^{ij},A_{ij})\big]$; the coefficients of $\log\varepsilon$ are logarithmic divergences, and $\log(s^2\varepsilon)=\log\varepsilon+2\log s$ produces the term $2\log s$ that carries the anomalous scaling.

Remark 1.3 (Physics notation: the Callan–Symanzik equation)**.** A tree insertion (Definition **??**) is the sum of the connected tree Feynman graphs with one insertion of F, one of X, any number of vertices of I, and internal lines $\hbar P(0,\infty)$, the unregulated Feynman-gauge propagators proportional to $1/|p|^2$. Theorem **??** is the one-loop Callan–Symanzik equation of the renormalised operator $O[\infty]$: $s\frac{d}{ds}\big|_{s=1}\sigma_s$ generates $x\mapsto sx$, $\big((E+w_O)O\big)[\infty]$ is its canonical part $x^m\partial_m+w_O$, and the two $\log\varepsilon$ terms are its anomalous part, the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon]$ being the one-loop renormalisation constant of O and the one in $I^{CT}[\varepsilon]$ the counterterm of the action inserted into $O[\infty]$. The factor 2 is that of $2\varepsilon\frac{d}{d\varepsilon}=-\frac{d}{d\log\varepsilon^{-1/2}}$, minus the logarithmic derivative with respect to the momentum cutoff; the power divergences ε^{-1} drop out.

Remark 1.4 (Physics notation: field-strength renormalisation of an operator)**.** The field-counting functional generates the one-loop wave-function renormalisation of Theorem **??**, a rescaling of A_m, A_I (weight 2), $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$ (weight 4) and c (weight 1) at fixed coupling (Corollary **??**). Relative to the kinetic terms $-\frac18(F_{mn},F_{mn})$, $-\frac{1}{16}(D_mA^{ij},D_mA_{ij})$, $i(\bar\lambda_{\dot\alpha i},\bar\sigma^{m\dot\alpha\alpha}D_m\lambda_\alpha{}^i)$, (D_mc,A^*_m) of Proposition **??**, the quadratic part of (**??**) has the factors 4, 4, 8, -1 in units, the sums $2+2$, $2+2$, $4+4$, $1-2$ of the weights. On a composite operator without antifields the bracket counts its fields with these weights: 4 for $\mathrm{tr}(\phi_I\phi_J)$ (two scalars, $\phi_I=A^a_IT^a$), 2 for the letters A_I and A_m, 4 for λ, $\bar\lambda$, 1 for c; it is the part of the renormalisation of O that comes from the wave-function renormalisation of its constituent fields.

Remark 1.5 (Physics notation: BRST variation and field redefinition)**.** $Q + \{I, -\} = \{S, -\}$ is the classical BV differential; on functionals of the fields alone it is the BRST variation, and $(Q+\{I,-\})O = 0$ is the BRST invariance of O, which holds for gauge-invariant composite operators built from F_{mn}, A_I, λ, $\bar\lambda$ and covariant derivatives; for them the terms built from $(Q + \{I, -\})O$ vanish. The bracket $\{\text{field-counting functional}, O\}$ is the change of O under the field rescaling of Lemma **??**. Corollary **??** splits the effect of the field-strength renormalisation on $O[\infty]$ into this rescaling of the fields in O, a BRST-exact (Q-exact) term, and terms proportional to the BRST variation of O.

Remark 1.6 (Physics notation: dilatation operator and anomalous dimensions)**.** The dilatation of physics, normalised by $[D, P_m] = P_m$, is $D = -E = -\frac{1}{2}\Delta$, the generator $-E$ of [SW, (3)]. Acting through the representation $x \mapsto -\Lambda^{(1)}(x)$ of Corollary **??**, it sends an operator with $|O| = 0$ to $(E+w_O+\text{coefficient})O$, modulo the terms built from $(Q+\{I,-\})O$: this is the physics form $[D, O(x)] = (x^m\partial_m + w_O + \gamma_O)O(x)$, with scaling dimension $w_O + \gamma_O$ and anomalous dimension γ_O equal to the coefficient of the table. The Δ-component $2(E + w_O + \gamma_O)O[\infty]$ multiplies an operator at the origin by $2(w_O + \gamma_O)$. The rule "coefficient $=$ 2 (third column $-$ fourth column)" is the decomposition of the anomalous dimension of a composite operator into its vertex part, twice the $\log\varepsilon$ coefficient of its renormalisation constant, plus the anomalous dimensions -4, -4, -8, -2 (in units) of its constituent fields A_I, A_m, λ, c. With the unit $-\frac{g^2N}{32\pi^2}$ of Remark **??**,

$$\gamma = 2\cdot 4 - 4 - 4 = 0 \quad (\text{traceless part, the } \mathbf{20'}),$$
$$\gamma = 2\cdot(-8) - 4 - 4 = -24 \text{ units} = \frac{3g^2N}{4\pi^2} \quad (\text{trace, the Konishi operator}).$$

Remark 1.7 (Physics notation: the Konishi operator and the $\mathbf{20'}$ as a mixing matrix)**.** With $\phi_I = A_I^a T^a$, $\phi_{ij} = \phi_I \Sigma^I_{ij}$, $\phi^{ij} = \frac{1}{2}\varepsilon^{ijkl}\phi_{kl}$ as after Definition **??**, the trace bilinear is the Konishi operator $4\hbar\int\omega\,\mathrm{tr}(\phi_I\phi_I) = \hbar\int\omega\,\mathrm{tr}(\phi^{ij}\phi_{ij})$, the superconformal primary of the long Konishi multiplet, and the traceless part is the half-BPS $\mathbf{20'}$, $\mathrm{tr}(\phi_I\phi_J) - \frac{1}{6}\delta_{IJ}\,\mathrm{tr}(\phi_K\phi_K)$. By linearity the coefficient row of the table of Example **??**, $-4(w_1, w_2)$ times the trace, is on the twenty-one bilinears $\mathrm{tr}(\phi_I\phi_J)$ the one-loop dilatation operator

$$\mathrm{tr}(\phi_I\phi_J) \longmapsto \frac{g^2N}{8\pi^2}\,\delta_{IJ}\,\mathrm{tr}(\phi_K\phi_K),$$

with eigenvalue 0 on the $\mathbf{20'}$ and $6\cdot\frac{g^2N}{8\pi^2} = \frac{3g^2N}{4\pi^2}$ on the Konishi operator.

Remark 1.8 (Physics notation: the letters)**.** The letters are the elementary fields smeared with ω: $\int\omega\,(X, \iota_w A) = w_I\int\omega\,(X, A_I)$, $\int\omega\,(X, \iota_v(A|_V)) = v_m\int\omega\,(X, A_m)$, $\int\omega\,(\zeta,\lambda) = \int\omega\big[(\lambda_\alpha{}^i, \zeta^\alpha{}_i) - (\bar\lambda_{\dot\alpha i}, \bar\zeta^{\dot\alpha i})\big]$ and $\int\omega\,(X, c)$; for $\mathfrak{su}(N)$ and $X = -igX^aT^a$, $(X, A_I) = 2\hbar X^a A^a_I$ and $(X, A_m) = 2\hbar X^a A^a_m$. Their coefficients, with the unit $-\frac{g^2N}{32\pi^2}$, are the one-loop anomalous dimensions of the elementary fields in Feynman gauge, gauge-dependent quantities:

$$\gamma_{A_I} = \frac{g^2N}{8\pi^2}, \qquad \gamma_{A_m} = \frac{g^2N}{8\pi^2}, \qquad \gamma_\lambda = \gamma_{\bar\lambda} = \frac{g^2N}{4\pi^2}, \qquad \gamma_c = \frac{g^2N}{16\pi^2},$$

γ_{A_m} modulo the terms built from $(Q + \{I, -\})O$ (Remark **??**). Each is -2 times the weight 2, 2, 4, 1 of the field in Lemma **??**, in units: the field-strength renormalisation of that field.

Remark 1.9 (Physics notation: the boundary graphs)**.** In momentum space L_ε has the components of $P(\varepsilon,\infty)$ with $e^{-\varepsilon|p|^2}/|p|^2$ replaced by $e^{-\varepsilon|p|^2}$, and

$$s\frac{d}{ds}\Big|_{s=1}\frac{e^{-s^2\varepsilon|p|^2}}{|p|^2} = 2\varepsilon\frac{d}{d\varepsilon}\frac{e^{-\varepsilon|p|^2}}{|p|^2} = -2\varepsilon\,e^{-\varepsilon|p|^2}:$$

an edge carrying $-2\varepsilon L_\varepsilon$ is the cutoff derivative of the regulated propagator, the kernel of the exact (Wilson–Polchinski) renormalisation-group equation. In Feynman-graph language, with two external scalars: graph (e) is the bubble through the scalar potential, graph (f) the triangle with a gluon A_m exchanged between the two scalar lines, graphs (c), (d) the mixing of the bilinear into $(A|_V, A|_V) = 2\hbar A^a_m A^a_m$, graph (a) the self-contraction of O, its mixing with the identity operator; the legs are the one-loop self-energies of the external scalars, the vertex function $2|p|^2(w,w')(Z,Z')$ being the scalar kinetic term in momentum space.

Remark 1.10 (Physics notation: the dilatation Ward identity)**.** $C^\bullet(\mathfrak{psl}(4|4))$ adds constant ghosts for the superconformal generators, d_{CE} is their BRST differential, and $[(d_{CE}+Q)O[\infty]](\Delta)$ is the quantum dilatation transformation of the renormalised operator, the content of its dilatation Ward identity. In the components of the key the vertex $S^{(1)}(\Delta)$ is the antifield coupling of the classical scale transformation $2x\cdot\partial + 2w$ of the fields of canonical dimension w:

$$S^{(1)}(\Delta) = \int \big\{ (2x\cdot\partial A_m + 2A_m, A^*_m) + \tfrac{1}{4}(2x\cdot\partial A_{ij} + 2A_{ij}, A^{*ij}) - (2x\cdot\partial \lambda_\alpha{}^i + 3\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) \\ + (2x\cdot\partial \bar\lambda_{\dot\alpha i} + 3\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}) - (2x\cdot\partial c, c^*)\big\}.$$

The graphs (a)–(y) are the one-loop graphs with one insertion of this vertex, joined to the rest through the heat-kernel edge K_ε; the equivariant master-equation defect along Δ is the breaking of the dilatation Ward identity by the regulator; Proposition **??** is the tree-level dilatation Ward identity.

Chapter 2

seven

Remark 2.1 (Physics notation: components as quantum transformations)**.** With the constant ghosts of $\mathfrak{psl}(4|4)$ added to the BRST differential, the component along x_1 is the quantum transformation of the renormalised operator O under the generator x_1, the one-loop form of $[x_1, O]$; at order $\hbar^0$ it is the classical variation $\{S^{(1)}(x_1), O\}$. For the odd generators its coefficients along $\xi_\alpha{}^i$, $\bar\xi_{\dot\alpha i}$, $\eta^\alpha{}_i$, $\bar\eta^{\dot\alpha i}$ are the transformations under Q^i_α, $\bar Q_{\dot\alpha i}$, $S^\alpha{}_i$, $\bar S^{\dot\alpha i}$ (Definition **??**). The parameter δ is the source of O, added to the action as $\delta\, O(\varepsilon)$. A local representative $\int \omega\, f_{x_1} = \int d^4x\, \omega(x) f_{x_1}(x)$ is the smeared composite operator $f_{x_1}(x)$, a polynomial in A_m, A_{ij}, $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$, c, the antifields and their derivatives at x.

Remark 2.2 (Physics notation: BRST-exactness of the components)**.** In the language of the BRST operator extended by constant ghosts (the extended Slavnov–Taylor operator of algebraic renormalisation), Lemma **??** states that the transformation of O under x_1 is the Q-variation of the part of $O[\infty]$ linear in the constant ghost of x_1. Its physical content is carried by the local composite operator $f_{x_1}(x)$, to which the selection rules of this chapter apply.

Remark 2.3 (Physics notation: conformal transformations of the fields)**.** On the fields, $v.$ is the two-component transformation of $S^{(1)}(v)$ written out after Definition **??**: the conformal transformation of the vector A_m, of the scalars A_{ij} of dimension 1, of the Weyl spinors $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$ of dimension $\frac{3}{2}$ and of the ghost c of dimension 0, with the spin term $-\frac{1}{8}X_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta$ on $\lambda_\alpha{}^i$ and the $SU(4)_R$ rotation $r^i{}_j = \frac{1}{4}R_{IJ}(\bar\Sigma^J\Sigma^I)^i{}_j$, traceless and anti-Hermitian for real R. Here $X_{mn} = \partial_m v_n - \partial_n v_m$ is the local rotation and $\frac{1}{4}\operatorname{div} v$ the local scale factor, $\partial_m v_n + \partial_n v_m = \frac{1}{2}\operatorname{div} v\, \delta_{mn}$ (the conformal Killing equation), and for A_m the weight term sits in $A_n\partial_m v^n$. The condition on the antifields makes A^*, λ^*, c^* primaries of dimension $4-w$, i.e. 3, $\frac{5}{2}$, 4 (Lemma **??**), the dimensions for which the field–antifield pairing, of total dimension 4, is invariant.

Remark 2.4 (Physics notation: the covariant generators)**.** The covariant generators are the translations P_b, the rotations $\mathfrak{so}(4) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ (the generators M, $\bar M$ of [SW, (8)]) and the R-symmetry $\mathfrak{su}(4)_R \cong \mathfrak{so}(6)_R$ (the generators R). They commute with the Feynman-gauge kinetic operator $H = -\partial^2$, with the gauge-fixing operator Q^{GF} and with the proper-time cutoff $e^{-\varepsilon|p|^2}$, so the regulated propagators and the counterterms are Poincaré and $SU(4)_R$ covariant. The spin representation $\frac{1}{4}\rho\circ q$ of a rotation acts on $\lambda_\alpha{}^i$ by $-\frac{1}{8}X_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta$, on $\bar\lambda_{\dot\alpha i}$ from the right by $+\frac{1}{8}X_{mn}(\bar\sigma^m\sigma^n)^{\dot\beta}{}_{\dot\alpha}$, and an R-rotation acts on the index of W by $r^i{}_j$.

Remark 2.5 (Physics notation: $\mathbf{20'}$ and Konishi mixing)**.** Under $SO(6)_R$, $\operatorname{Sym}^2\mathbf{6} = \mathbf{1}\oplus\mathbf{20'}$. With $\phi_I = A^a_I T^a$ and $(\iota_{w_1}A, \iota_{w_2}A) = 4\hbar\, w_{1I}w_{2J}\operatorname{tr}(\phi_I\phi_J)$, the $\mathbf{20'}$ operators $\operatorname{tr}(\phi_I\phi_J) - \frac{1}{6}\delta_{IJ}\operatorname{tr}(\phi_K\phi_K)$ renormalise multiplicatively, and the Konishi operator $\operatorname{tr}(\phi_I\phi_I)$ is renormalised into the span of itself, of the gauge-variant $(A|_V, A|_V) = 2\hbar A^a_m A^a_m$ and of the identity operator $\int\omega$. The coefficient of the identity, $\frac{2\hbar\dim\mathfrak{g}\,(w_1,w_2)}{16\pi^2\varepsilon}$ with $\dim\mathfrak{su}(N) = N^2-1$, is a power divergence, the subtraction of the vacuum expectation value of the bilinear; the coefficient of $(A|_V, A|_V)$ is 0.

Remark 2.6 (Physics notation: classical action of P, M, $\bar{M}$, R)**.** Remark **??** in physics language: for BRST-invariant O the one-loop transformations under P_b, M, $\bar{M}$ and R can be chosen to be the classical ones (derivative, rotation of the Lorentz indices, $SU(4)_R$ rotation of the W indices), with no anomalous term. The internal insertions exist because the gauge-fixing operator squares to a nonzero operator; in components

$$(Q^{GF})^2 A^*_m = -2\,\partial_m A^*_m \quad \text{(in the slot of } c\text{)},$$

the composition of $A^*_m \mapsto -2A^*_m$ (slot of A_m) with $A_m \mapsto \partial_m A_m$ (slot of c), equal to $2d^*(A^*|_V)$ with $d^* = -\partial_m$ on one-forms.

Remark 2.7 (Physics notation: dilatation and special conformal fields)**.** In components the dilatation is $v^m = 2x^m$, $\operatorname{div} v = 8$, and the special conformal field along b is

$$v^m = x^2 b^m - 2(b\cdot x)x^m, \qquad X_{mn} = 4(x_m b_n - x_n b_m), \qquad \operatorname{div} v = -8\, b\cdot x.$$

On the scalars $v.A_{ij} = \big(x^2 b^n - 2(b\cdot x)x^n\big)\partial_n A_{ij} - 2(b\cdot x)A_{ij}$, the special conformal transformation $b^n\big(x^2\partial_n - 2x_n\, x\cdot\partial - 2w\, x_n\big)$ of a scalar primary of dimension $w = 1$. The proposition states that Δ and K_b fail to commute with the Feynman-gauge operator Q^{GF}, and that the dilatation acts on the propagator as the rescaling of the proper-time cutoff: $(v.\otimes 1 + 1\otimes v.)P(\varepsilon,\infty) = -4\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty)$ is, on each component, $4\varepsilon\, e^{-\varepsilon|p|^2}$ times the numerator of the propagator.

Remark 2.8 (Physics notation: free-field supersymmetry and the gauge-fixing operator)**.** $Q_1 \in S_+\otimes W$ is a combination of the supercharges Q^i_α, in $(\mathbf{2},\mathbf{1},\mathbf{4})$, and $Q_1 \in S_-\otimes W^*$ of the $\bar{Q}_{\dot\alpha i}$, in $(\mathbf{1},\mathbf{2},\bar{\mathbf{4}})$. The field transformations of $S^{(1)}(Q_1)$ are the $\mathcal{N}=4$ supersymmetry transformations written out at the end of Section 2.3, $Q_1.A = -\Gamma(Q_1,\lambda)$, $Q_1.\lambda = -\frac{1}{2}\rho(F_A)Q_1$; the operator $Q_1.$ of the proposition is their free part ($F_{mn}\to\partial_m A_n - \partial_n A_m$, $D_m A^{ij}\to\partial_m A^{ij}$, the term $\frac{1}{4}[A^{ik},A_{kj}]\xi_\alpha{}^j$ dropped), and the cubic term of the last item consists of the commutator terms. The second and third items state that the regulated Feynman-gauge propagator is not invariant under the supercharges: the commutator with Q^{GF} has a component from A^* to λ, the sum of the compositions through A (the gauge-fixing arrow $A^*\mapsto -2A^*$) and through λ^* (the arrow $\lambda^*\mapsto -\not{\partial}|_{\Sigma^*}\lambda^*$).

Remark 2.9 (Physics notation: special supersymmetry and large distance)**.** $Q_2 \in \Sigma^*$ is a combination of the special supercharges $S^\alpha{}_i$, in $(\mathbf{2},\mathbf{1},\bar{\mathbf{4}})$, and $\bar{S}^{\dot\alpha i}$, in $(\mathbf{1},\mathbf{2},\mathbf{4})$. The multiplication operator $x\mapsto\Gamma(\rho(x)Q_2,-)$ is that of the conformal Killing spinor $\epsilon_\alpha{}^i(x) = -i\,x_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, $\bar\epsilon_{\dot\alpha i}(x) = i\,x_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$, linear in x: the special supersymmetry transformations of the fields are position dependent, and this multiplication operator fails to commute with $H = -\partial^2$. In Remark **??** the coefficients grow as $|x|$ for Δ and for $S^\alpha{}_i$, $\bar{S}^{\dot\alpha i}$, and as $|x|^2$ for K_b; the decay $|y|^{-4}$ of a K_b insertion is the borderline of a logarithmic infrared divergence.

Remark 2.10 (Physics notation: dimensional analysis of the components)**.** The proposition is dimensional analysis together with ghost-number conservation, gauge invariance and $SO(4)\times SU(4)_R$ covariance. The value of the conformal Killing spinor (ξ at $x_0 = 0$) has mass dimension $-\frac{1}{2}$ and its Dirac derivative (4η) dimension $+\frac{1}{2}$, so the transformation under Q has dimension $w_O + \frac{1}{2}$ and the η-term dimension $w_O - \frac{1}{2}$; a translation parameter has dimension -1, rotation and dilatation parameters 0, and the special conformal parameter b dimension $+1$. In components the conformal Killing spinor (**??**) satisfies

$$\partial_m\epsilon_\alpha{}^i = -i\,\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}, \qquad \partial_m\bar\epsilon_{\dot\alpha i} = i\,\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}, \qquad i\,\partial_m\bar\epsilon_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha} = 4\eta^\alpha{}_i, \qquad -i\,\bar\sigma^{m\dot\alpha\alpha}\partial_m\epsilon_\alpha{}^i = 4\bar\eta^{\dot\alpha i},$$

the last two being the components of $\not{\partial}(Q_1 + \rho(x)Q_2) = 4Q_2$.

Remark 2.11 (Physics notation: structures of the one-loop supersymmetry variation)**.** For $O = 4\hbar\, w_{1I} w_{2J} \int \omega\, \mathrm{tr}(\phi_I \phi_J)$ the one-loop transformation under Q is a dimension-$\frac{5}{2}$ combination of structures with the fields $A_I \lambda$ (classical type), $A_m \lambda$ (gauge-variant type) and $c\lambda^*$ (BRST type). In two-component form, with the colour pairing between the two $\mathfrak{g}$-valued factors, for Q^i_α, $Q_1 = (\xi_\alpha{}^i, 0)$,

$$(\rho(w)Q_1, \lambda) = w_{ji}\, \xi_\alpha{}^j \varepsilon^{\alpha\beta} \lambda_\beta{}^i, \qquad (\rho(v)Q_1, \lambda) = i\, v_m \bar\lambda_{\dot\alpha i} \bar\sigma^{m\dot\alpha\alpha} \xi_\alpha{}^i, \qquad (Q_1, \lambda^*) = \xi_\alpha{}^i \lambda^{*\alpha}{}_i,$$

and for $\bar Q_{\dot\alpha i}$, $Q_1 = (0, \bar\xi_{\dot\alpha i})$,

$$(\rho(w)Q_1, \lambda) = -w^{kl}\, \bar\xi_{\dot\alpha k} \varepsilon^{\dot\alpha\dot\beta} \bar\lambda_{\dot\beta l}, \qquad (\rho(v)Q_1, \lambda) = i\, v_m \bar\xi_{\dot\alpha i} \bar\sigma^{m\dot\alpha\alpha} \lambda_\alpha{}^i, \qquad (Q_1, \lambda^*) = -\bar\xi_{\dot\alpha i} \bar\lambda^{*\dot\alpha i},$$

with $w_{ji} \to A_{ji}$, $w^{kl} \to A^{kl}$ for $\rho(A|_{W_{\mathbb{R}}^\perp})$ and $v_m \to A_m$ for $\rho(A|_V)$; $(\rho(w)Q_1, \lambda) = \iota_w \Gamma(Q_1, \lambda) = \frac{1}{4} w^{kl} \Gamma_{kl}(Q_1, \lambda)$. In $SU(4)_R$ language, $\mathbf{4} \otimes \bar{\mathbf{4}} = \mathbf{1} \oplus \mathbf{15}$, $\wedge^2 \mathbf{6} = \mathbf{15}$ (the adjoint) and $\mathrm{Sym}^2 \mathbf{6} = \mathbf{1} \oplus \mathbf{20'}$, and $\mathbf{20'}$ does not occur in $\mathbf{1} \oplus \mathbf{15}$.

Remark 2.12 (Physics notation: special supercharges at a point)**.** For $S^\alpha{}_i$ the Killing spinor at x is $\bar\epsilon_{\dot\alpha i}(x) = i\, x_m \eta^\alpha{}_i \sigma^m_{\alpha\dot\alpha} \in S_- \otimes W^*$, and for $\bar S^{\dot\alpha i}$ it is $\epsilon_\alpha{}^i(x) = -i\, x_m \sigma^m_{\alpha\dot\alpha} \bar\eta^{\dot\alpha i} \in S_+ \otimes W$. At the position x of the operator the one-loop transformation under $S^\alpha{}_i$ therefore has the structures of the $\bar Q_{\dot\alpha i}$-variation with parameter $i\, x_m \eta^\alpha{}_i \sigma^m_{\alpha\dot\alpha}$, and the one under $\bar S^{\dot\alpha i}$ the structures of the Q^i_α-variation with parameter $-i\, x_m \sigma^m_{\alpha\dot\alpha} \bar\eta^{\dot\alpha i}$, with coefficients of their own.

Remark 2.13 (Physics notation: special conformal transformations at a point)**.** For a scalar primary of dimension w the classical special conformal transformation is $b^m\big(x^2 \partial_m - 2x_m\, x \cdot \partial - 2w\, x_m\big) = v^m \partial_m + \frac{w}{4} \operatorname{div} v$. At x_0 its data are the translation $v(x_0)$, the local dilatation $\frac{1}{4} \operatorname{div} v(x_0) = -2\, b \cdot x_0$, the local rotation $X_{mn}(x_0) = 4(x_{0m} b_n - x_{0n} b_m)$ and the constant second derivatives, linear in b. The conformal expectation of Remark **??** is the physics rule that a primary of dimension $w_O + \gamma$ transforms with $-2(w_O + \gamma)\, b \cdot x_0$, the anomalous dimension γ of Theorem **??** entering through the local dilatation.

Remark 2.14 (Physics notation: the generators and their dimensions)**.** With the dilatation of physics $D = -E = -\frac{1}{2}\Delta$ (the generator $-E$ of [SW, (3)]), the column ad_Δ gives

$$\begin{aligned} &[D, P_b] = P_b, \qquad [D, Q^i_\alpha] = \tfrac{1}{2} Q^i_\alpha, \qquad [D, \bar Q_{\dot\alpha i}] = \tfrac{1}{2} \bar Q_{\dot\alpha i}, \\ &[D, S^\alpha{}_i] = -\tfrac{1}{2} S^\alpha{}_i, \qquad [D, \bar S^{\dot\alpha i}] = -\tfrac{1}{2} \bar S^{\dot\alpha i}, \qquad [D, K_b] = -K_b, \end{aligned}$$

the dimensions 1, $\frac{1}{2}$, $-\frac{1}{2}$, -1 carried by P, Q, S, K, while M, $\bar M$, R and D have dimension 0. The column "weight of output" is the canonical dimension of the transformed operator, the dimension of O plus that of the generator, and for $S^\alpha{}_i$, $\bar S^{\dot\alpha i}$ evaluated on the Killing spinor $\rho(x)Q_2$ of dimension $-\frac{1}{2}$.

Remark 2.15 (Physics notation: anomalies and finite counterterms)**.** (**??**) is the breaking, by the regulator, of the quantum master equation, i.e. of the Slavnov–Taylor identity of BRST invariance (arity zero) and of the Ward identities of the superconformal generators (arity one); its flow to scale ∞ is the one-loop obstruction, a potential anomaly, and Step 2 is the algebraic-renormalisation problem of removing it by finite local counterterms. The finite counterterm of Chapter 4, the Feynman-gauge longitudinal term, reads for $\mathfrak{su}(N)$, with $d^* A|_V = -\partial_m A_m$,

$$\frac{27}{16} \frac{\hbar\, \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X, Y)} \int d^4x\, (\partial_m A_m, \partial_n A_n) = -\frac{27 g^2 N}{256\pi^2}\, \hbar \int d^4x\, (\partial_m A^a_m)^2.$$

Remark 2.16 (Physics notation: Step 3)**.** Step 3 computes the one-loop corrections to the transformations of the letters and of the bilinears $\mathrm{tr}(\phi_I \phi_J)$ under Q^i_α, $\bar Q_{\dot\alpha i}$, $S^\alpha{}_i$, $\bar S^{\dot\alpha i}$: one number per generator for the half-BPS $\mathbf{20'}$, three for the Konishi operator $\mathrm{tr}(\phi_I \phi_I)$, the superconformal primary of the long Konishi multiplet. $S^{(2)}$ is quadratic in the gaugino antifields and contains the coupling of $\Gamma^m(\epsilon_1, \epsilon_2) A_m + \frac{1}{4} \Gamma_{ij}(\epsilon_1, \epsilon_2) A^{ij}$ to c^*: in physics language the supersymmetry algebra closes on the fields only up to a field-dependent gauge transformation and the gaugino equation of motion, and the arity-two components express this on-shell closure at one loop.

Remark 2.17 (Physics notation: Step 4)**.** The letters are the scalars A_I (in $SU(4)$ form A_{ij}), the field strength F_A with components F_{mn}, $D_m A_I$, $[A_I, A_J]$, the gauginos $\lambda_\alpha{}^i$, $\bar{\lambda}_{\dot{\alpha} i}$, and their covariant derivatives $D_m = \partial_m + [A_m, -]$. Step 4 is the complete one-loop mixing matrix (the one-loop dilatation operator) on gauge-invariant two-letter operators, in blocks of fixed canonical dimension, ghost number and $SO(4) \times SU(4)_R$ representation. The superficial degree is the power counting of physics: the dimension of the vertex O minus the canonical dimensions of the external lines.

Remark 2.18 (Physics notation: Step 5)**.** Order $\hbar^2$ is two loops, order g^4 in the normalisation of Remark **??**. The one-loop counterterm vertices inserted into one-loop graphs subtract the subdivergences, $(\log \varepsilon)^2$ is the leading double logarithm of the proper-time cutoff, and the two-loop dilatation read from the $\log \varepsilon$ coefficients is the two-loop anomalous-dimension matrix.

Chapter 3

eight

Remark 3.1 (Physics notation: ten-dimensional and $SO(6)$ gamma matrices)**.** With $\zeta_A = (\xi_\alpha{}^i, \bar\xi_{\dot\alpha i}) \in \Sigma = \mathbf{16}$ and $\chi^A = (\eta^\alpha{}_i, \bar\eta^{\dot\alpha i}) \in \Sigma^* = \mathbf{16'}$ of Spin(10) (the key after Section 1.2), the gamma matrices $\Gamma_m = \rho(e_m)$ are the formulas of the definition at $v = e_m$, and the six transverse ones $\Gamma_I = \rho(e_I)$ are, with $\bar\Sigma^I = (\Sigma^I)^\dagger$ and $w^{ij} = -w_I \bar\Sigma^{I\,ij}$,

$$(\Gamma_I\zeta)^\alpha{}_j = \varepsilon^{\alpha\beta}\Sigma^I_{ji}\xi_\beta{}^i, \qquad (\Gamma_I\zeta)^{\dot\alpha j} = -\varepsilon^{\dot\alpha\dot\beta}\bar\Sigma^{I\,ji}\bar\xi_{\dot\beta i},$$
$$(\Gamma_I\chi)_\alpha{}^k = \bar\Sigma^{I\,kj}\varepsilon_{\alpha\beta}\eta^\beta{}_j, \qquad (\Gamma_I\chi)_{\dot\beta k} = -\Sigma^I_{kj}\varepsilon_{\dot\beta\dot\alpha}\bar\eta^{\dot\alpha j}.$$

The 't Hooft blocks are the chiral blocks of the $SO(6)_R$ gamma matrices: on $W \oplus W^*$,

$$\gamma^I = \begin{pmatrix} 0 & \bar\Sigma^I \\ \Sigma^I & 0 \end{pmatrix}, \qquad \gamma^I\gamma^J + \gamma^J\gamma^I = 2\delta^{IJ}, \qquad (\gamma^I)^\dagger = \gamma^I,$$

so $W = \mathbf{4}$ and $W^* = \bar{\mathbf{4}}$ are the two Weyl spinors of Spin(6) $\cong SU(4)_R$, and $W^\perp_{\mathbb{R}}$ is its vector $\mathbf{6}$. The generators Q^i_α, $\bar Q_{\dot\alpha i}$, $S^\alpha{}_i$, $\bar S^{\dot\alpha i}$ span $(\mathbf{2},\mathbf{1},\mathbf{4})$, $(\mathbf{1},\mathbf{2},\bar{\mathbf{4}})$, $(\mathbf{2},\mathbf{1},\bar{\mathbf{4}})$, $(\mathbf{1},\mathbf{2},\mathbf{4})$ of $SU(2)_L \times SU(2)_R \times SU(4)_R$.

Remark 3.2 (Physics notation: Feynman gauge with two cutoffs)**.** The propagator is that of Feynman gauge. As in Proposition **??**, $\hbar$ times the inverse pairing is $\frac12\delta^{ab}$ in the basis $-igT^a$ and δ^{ab} in the basis $-\frac{ig}{\sqrt2}T^a$, so the edges $\hbar P(\varepsilon, T)$ of the flow are the canonical Euclidean propagators

$$\hbar\langle A^a_m A^b_n\rangle = \delta^{ab}\delta_{mn}, \qquad \hbar\langle A^a_I A^b_J\rangle = \delta^{ab}\delta_{IJ}, \qquad \hbar\langle \lambda^{a\,i}_\alpha\, \bar\lambda^b_{\dot\beta j}\rangle = -\delta^{ab}\delta^i_j\, p_m\sigma^m_{\alpha\dot\beta},$$

each times $\big(e^{-\varepsilon|p|^2} - e^{-T|p|^2}\big)/|p|^2$, with the proper-time cutoffs ε (ultraviolet) and T (infrared). In $SO(6)$ form, $\langle A_{ij}A_{kl}\rangle = 4(-,-)^{-1}\varepsilon_{ijkl}$ is $\langle A_I A_J\rangle = 2(-,-)^{-1}\delta_{IJ}$, by $A_I = \frac14\bar\Sigma^{I\,ji}A_{ij}$.

Remark 3.3 (Physics notation: odd brackets on unit elements)**.** On the unit elements of Definition **??**, the odd–odd brackets of Chapter 2 (graded brackets, $P_m = \partial_m$) are

$$[Q^i_\alpha, \bar Q_{\dot\alpha j}] = i\,\delta^i_j\,\bar\sigma^{m\dot\alpha\alpha}P_m, \qquad [Q^i_\alpha, Q^j_\beta] = 0, \qquad [\bar Q_{\dot\alpha i}, \bar Q_{\dot\beta j}] = 0,$$

the values of $[\xi_1,\xi_2] = i\big(\bar\xi_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{2\alpha}{}^i + \bar\xi_{2\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{1\alpha}{}^i\big)P_m$, the Euclidean form of the anticommutator of Q and $\bar Q$ into a translation ($\bar\sigma^m P_m = \bar\sigma^\mu P_\mu$ after the Wick map). The dilatation part of $[Q^i_\alpha, S^\beta{}_j]$ is $\frac12\delta_\alpha{}^\beta\delta^i{}_j\Delta$ and that of $[\bar Q_{\dot\alpha i}, \bar S^{\dot\beta j}]$ is $-\frac12\delta_{\dot\alpha}{}^{\dot\beta}\delta_i{}^j\Delta$, from the term $\frac12(\xi,\eta)\Delta$ of $[\xi,\rho(x)\eta]$; the brackets of Q with $\bar S$ and of $\bar Q$ with S have no dilatation part.

Remark 3.4 (Physics notation: infrared endpoint and Wilsonian flow)**.** T is an infrared proper-time cutoff, of length $T^{1/2}$, and ε the ultraviolet one: the scalar factor of $P(\varepsilon, T)$ is the massless propagator

$\frac{1}{4\pi^2|x-y|^2}$ with both cutoffs, $\frac{e^{-|x-y|^2/4T}-e^{-|x-y|^2/4\varepsilon}}{4\pi^2|x-y|^2}$. $W(P(\varepsilon,T),J)$ is the Wilsonian effective interaction at scale T, the connected Feynman graphs with vertices from $J/\hbar$ and internal lines $\hbar P(\varepsilon,T)$, in which the modes of proper time between ε and T are integrated out. Lemma **??** is the semigroup property of this exact renormalisation-group flow and its compatibility with the BV differential.

Remark 3.5 (Physics notation: the equivariant quantum master equation)**.** $I_{\mathfrak{psl}(4|4)}[T]$ is the Wilsonian effective action at scale T with the constant superconformal ghosts. (**??**) is the regularised Slavnov–Taylor identity: its arity-zero part is that of BRST invariance, its arity-one part along x the Ward identity of the generator x, its arity-two part the closure of the algebra. $F_{\mathfrak{psl}(4|4)}$ is a finite symmetry-restoring counterterm in the sense of algebraic renormalisation, and the functions $\varepsilon^{-m}\log^k\varepsilon$ are the power and logarithmic divergences of the proper-time cutoff.

Remark 3.6 (Physics notation: anomaly and renormalisation ambiguities)**.** Ob is the one-loop anomaly of the regularised Slavnov–Taylor and superconformal Ward identities; (i) is its Wess–Zumino consistency condition, (ii) states that it is removed by a finite counterterm exactly when it is s-exact, and (iii) identifies the finite renormalisation ambiguities with H^0 of s on local cochains, the field redefinitions being trivial. In (iv) the coupling direction is a finite renormalisation of the coupling: through the pairing $(X,Y)=-\frac{4\hbar}{g^2}\operatorname{tr}(XY)$ the action S is proportional to g^{-2} at fixed $\mathfrak{g}$-valued fields, and μS is S with $g^{-2}\mapsto\mu g^{-2}$. G_B is the relative normalisation of Q and $\bar{Q}$ (Theorem **??** (ii)).

Remark 3.7 (Physics notation: the hypercharge)**.** B is the $\mathfrak{u}(1)$ outer automorphism of $\mathfrak{psu}(2,2|4)$, the hypercharge $U(1)_Y$: in $\mathfrak{gl}(4|4)$, with real form $\mathfrak{u}(2,2|4)$, it is the commutator with $\operatorname{diag}(0_4,1_4)$. It vanishes on P, M, $\bar{M}$, Δ, K, R and counts upper minus lower $SU(4)_R$ indices on the supercharges,

$$B(Q^i_\alpha)=Q^i_\alpha,\qquad B(\bar{Q}_{\dot\alpha i})=-\bar{Q}_{\dot\alpha i},\qquad B(S^\alpha{}_i)=-S^\alpha{}_i,\qquad B(\bar{S}^{\dot\alpha i})=\bar{S}^{\dot\alpha i},$$

so Q^i_α and $\bar{S}^{\dot\alpha i}$ (in the $\mathbf{4}$) have charge $+1$ and $\bar{Q}_{\dot\alpha i}$ and $S^\alpha{}_i$ (in the $\bar{\mathbf{4}}$) charge -1. In components $G^{(1)}_B(x)=S^{(1)}(Bx)$ is $S^{(1)}$ with $\xi_\alpha{}^i$, $\bar\eta^{\dot\alpha i}$ kept and $\bar\xi_{\dot\alpha i}$, $\eta^\alpha{}_i$ reversed in sign.

Remark 3.8 (Physics notation: the two finite ambiguities)**.** e^{tB} multiplies Q^i_α, $\bar{S}^{\dot\alpha i}$ by e^t and $\bar{Q}_{\dot\alpha i}$, $S^\alpha{}_i$ by e^{-t} and fixes the even part, an automorphism of the superconformal algebra; G_B is the corresponding change of the couplings to the constant ghosts, with $G^{(0)}_B=0$, so the action S itself is unchanged. Proposition **??** states that this ambiguity, a relative rescaling of Q and $\bar{Q}$, is independent of the finite renormalisation of the coupling.

Remark 3.9 (Physics notation: the one-loop generators)**.** $Q_1(O)^\alpha{}_i$, $\bar{Q}_1(O)^{\dot\alpha i}$, $S_1(O)_\alpha{}^i$, $\bar{S}_1(O)_{\dot\alpha i}$ are the one-loop corrections to the transformations of O under Q^i_α, $\bar{Q}_{\dot\alpha i}$, $S^\alpha{}_i$, $\bar{S}^{\dot\alpha i}$; P_1, M_1, R_1, K_1, D_1 are those under P_b, $\mathfrak{so}(4)$, $\mathfrak{su}(4)_R$, K_b, Δ; $\hbar L_1(O)^{(0)}$ is the one-loop correction to the BRST variation of O. For the dilatation $D=-\frac12\Delta$ of physics, acting through $-\Lambda^{(1)}$ (Corollary **??**), the one-loop dilatation operator is $\frac12 D_1$: $\frac12 D_1(O)=-24$ units times O, i.e. $\frac{3g^2N}{4\pi^2}O$, for the Konishi operator, and 0 on the $\mathbf{20'}$ (Remark **??**).

Remark 3.10 (Physics notation: renormalised operators and their transformations)**.** $M(\varepsilon)$ is the bare operator with its counterterms, ψ the finite part that fixes the renormalisation scheme of the composite operators ($\psi=0$ is minimal subtraction of the singular terms $\varepsilon^{-m}\log^k\varepsilon$), and $M[T]$ the renormalised operator in the Wilsonian theory at scale T. d_T is the quantum BRST differential at scale T, and $L(M)$ the local composite operator representing the quantum transformation of M, unique by Theorem **??** (i). The cohomology of $\Lambda^{(0)}$ on local functionals is the space of physical local operators (BRST-closed modulo BRST-exact), and by Corollary **??** $x\mapsto-\Lambda^{(1)}(x)$ is the action of the superconformal algebra on it, closed up to BRST-exact terms; in this normalisation $D=-\frac12\Delta$ acts on an operator of degree 0 at the origin as multiplication by its scaling dimension $w_O+\gamma_O$.

Remark 3.11 (Physics notation: scheme dependence)**.** (a) and (b) are field redefinitions and finite renormalisations $O \mapsto O + \hbar\psi'(O)$ of the composite operators, changes of scheme that conjugate all one-loop generators. (c) with the coupling direction is a finite renormalisation of g, which rescales the one-loop BRST operator, and with G_B the hypercharge rescaling of the supercharges, Q^i_α and $\bar{S}^{\dot\alpha i}$ by $1-a\hbar$, $\bar{Q}_{\dot\alpha i}$ and $S^\alpha{}_i$ by $1+a\hbar$. Theorem **??** (iii) is the scheme independence of the one-loop anomalous dimensions, the eigenvalues of $\frac{1}{2}D_1$ on operators of fixed canonical dimension.

Remark 3.12 (Physics notation: the twisting supercharge)**.** $[\xi,\xi]=0$ for $\xi \in S_+ \otimes W$ is $[Q^i_\alpha, Q^j_\beta]=0$. For $\xi = u\otimes z$ the ten-dimensional spinor $\zeta_A = (u_\alpha z^i, 0)$ is a pure spinor, $\Gamma^M(\xi,\xi)=0$ for all ten $M=(m,I)$. For $z^i=\delta^i{}_4$ and u_α a weight vector of $\mathfrak{su}(2)_L$, ξ is the single supercharge written Q^4_- in the physics literature on the holomorphic twist; every nonzero $u\otimes z$ is of this form after an $SU(2)_L\times SU(4)_R$ rotation and a rescaling. r_ξ sets the constant ghost of ξ equal to q, then to 1: S^ξ is the BV action of the theory twisted by Q^4_-, its differential $\{S^\xi,-\}$ is BRST plus Q^4_-, and

$$-\tfrac{1}{2}S^{(2)}(\xi,\xi) = \tfrac{1}{4}\int \left(u_\alpha z^i \lambda^{*\alpha}{}_i,\ u_\beta z^j \lambda^{*\beta}{}_j\right)$$

is the term quadratic in the gaugino antifields that is required because $\{S^{(1)}(\xi), S^{(1)}(\xi)\} = \{S, S^{(2)}(\xi,\xi)\}$ is a nonzero multiple of the gaugino equation of motion: Q^4_- squares to zero on shell.

Remark 3.13 (Physics notation: the twisted one-loop differential)**.** $Q^\xi_0 = \{S^\xi,-\}$ is the classical twisted differential, BRST plus Q^4_-; its cohomology on local operators is the classical Q^4_--cohomology, the $\frac{1}{16}$-BPS sector of the physics literature. $\hbar Q^\xi_1$ is its one-loop correction, the one-loop BRST correction $\hbar L^{(0)}_1$ plus the one-loop Q^4_--component Q_1 plus the arity-two terms, and $[Q^\xi_1]$ is the scheme-independent one-loop correction of Q^4_- on this cohomology, the object compared with the one-loop differential of [BK].

Remark 3.14 (Physics notation: infrared behaviour)**.** The endpoint T is an infrared regulator. The insertions of $S^{(1)}$ along Q, S, Δ, P, M, $\bar{M}$, R are infrared finite, and the special conformal K_b, whose Killing field $x^2 b^m - 2(b\cdot x)x^m$ grows quadratically, produces an infrared logarithm, $\int_{|y|^2<T}|y|^{-4}d^4y$ growing as $\pi^2\log T$. Theorem **??** states that the local operator $K_1(O)$ is nevertheless infrared safe, the $\log T$ residing in the Wilsonian flow of $O[T]$; in radial quantisation K_b is the conjugate of P_b (Corollary **??**).

Remark 3.15 (Physics notation: Euclidean conjugation of spinors)**.** In Euclidean signature complex conjugation keeps the four-dimensional chirality and exchanges **4** and $\bar{\mathbf{4}}$: $\overline{\xi_\alpha{}^i}$, placed as $\eta^\alpha{}_i$, lies in $(\mathbf{2},\mathbf{1},\bar{\mathbf{4}})$, and $-\overline{\bar\xi_{\dot\alpha i}}$, placed as $\bar\eta^{\dot\alpha i}$, in $(\mathbf{1},\mathbf{2},\mathbf{4})$; upper and lower indices are exchanged as for the unitary groups $SU(2)_L$, $SU(2)_R$, $SU(4)_R$ of the compact real form. In Lorentzian signature, by contrast, $(\psi_\alpha)^\dagger$ is a dotted spinor. In ten-dimensional notation the conjugation maps $\zeta_A \in \mathbf{16}$ to an element of $\mathbf{16'}$, with upper index, compatibly with the gamma matrices: $\overline{\Gamma_M\zeta} = \Gamma_M\bar\zeta$ for the real basis vectors e_M.

Remark 3.16 (Physics notation: radial quantisation)**.** This is the conjugation of radial quantisation, $P^\dagger = K$, $Q^\dagger = S$, $\bar{Q}^\dagger = \bar{S}$, $D^\dagger = D$ in physics notation, here with the signs $P_b^\dagger = -K_{\bar b}$, $(Q^i_\alpha)^\dagger = -S^\alpha{}_i$, $(\bar{Q}_{\dot\alpha i})^\dagger = \bar{S}^{\dot\alpha i}$, $\Delta^\dagger = \Delta$, and with the matrix adjoint on $\mathfrak{su}(4)_R$. By the dilatation part $\frac{1}{2}(\xi,\eta)\Delta$ of $[\xi,\rho(x)\eta]$, both $[Q^i_\alpha,(Q^i_\alpha)^\dagger]$ and $[\bar{Q}_{\dot\alpha i},(\bar{Q}_{\dot\alpha i})^\dagger]$ (no sum) contain $-\frac{1}{2}\Delta = D$, the physics dilatation; (iii) states that on a superconformal primary (annihilated by Σ^*, i.e. by S and $\bar{S}$) of scaling dimension $d>0$ the norms of its Q-descendants are positive for the sign $+$. The real form $\mathfrak{psu}(2,2|4)$ has even part $\mathfrak{su}(2,2)\oplus\mathfrak{su}(4)_R$, with $\mathfrak{su}(2,2)\cong\mathfrak{so}(4,2)$ the Lorentzian conformal algebra, and maximal compact even subalgebra $\mathfrak{su}(2)_L\oplus\mathfrak{su}(2)_R\oplus\mathfrak{u}(1)\oplus\mathfrak{su}(4)_R$, the $\mathfrak{u}(1)$ generated by $i\Delta$.

Remark 3.17 (Physics notation: Osterwalder–Schrader reflection)**.** Θ_n is the Osterwalder–Schrader reflection in the hyperplane orthogonal to n; for $n=e_4$ it is Euclidean time reflection $x^4\mapsto -x^4$, and

$\dagger_n$ corresponds to the Hermitian conjugation of the Wick-rotated Lorentzian algebra. On the scalars it uses the reality of the **6**, $\overline{A_{ij}} = A^{ij}$, and on the gauginos it exchanges λ and $\bar\lambda$. For $n = e_4$, where $\sigma^4 = i$,

$$(Q^i_\alpha)^{\dagger_{e_4}} = -\bar Q_{\dot\alpha i}\big|_{\dot\alpha=\alpha}, \qquad (\bar Q_{\dot\alpha i})^{\dagger_{e_4}} = -Q^i_\alpha\big|_{\alpha=\dot\alpha},$$

the Euclidean form of the Lorentzian relation $(Q_\alpha)^\dagger = \bar Q_{\dot\alpha}$ between undotted and dotted supercharges. The reflection inner product $\langle a, b\rangle_n$ of Definition **??** is the Osterwalder–Schrader inner product.

Remark 3.18 (Physics notation: BPZ conjugation)**.** C is the conformal map from the half space to the unit ball that carries planar (Osterwalder–Schrader) quantisation to radial quantisation, and Θ_I, the inversion $x \mapsto x/x^2$ with the conformal factors, is the BPZ conjugation of radial quantisation. Proposition **??** is the BPZ conjugate of a scalar primary of scaling dimension $w_O+\kappa$, with κ its one-loop anomalous dimension ($D_1(O) = 2\kappa O$),

$$O(x)^\dagger = |x|^{-2(w_O+\kappa)}\,\overline{O}\big(x/|x|^2\big),$$

the anomalous part $K_1(O) = -2(b\cdot x)\kappa O$ of the special conformal transformation being that of $b^m\big(x^2\partial_m - 2x_m\, x\cdot\partial - 2(w_O+\kappa)x_m\big)$.

Remark 3.19 (Physics notation: two-point functions)**.** (i) is the decoupling of BRST-exact operators, (ii) the absence of ultraviolet and infrared problems at separated points, and (iii) the Callan–Symanzik equation of the two-point function, with ℓ the renormalisation length. For a single scalar operator with $D_1(O) = 2\gamma O$ ($\gamma = \frac{3g^2N}{4\pi^2}$ for the Konishi operator, 0 for the **20′**) it reads

$$\langle O, O\rangle(x,0) = |x|^{-2w}\Big[C + (\text{one-loop finite part}) - \gamma\, C \log\frac{|x|^2}{\ell^2}\Big],$$

the one-loop expansion of $C\,(|x|^2/\ell^2)^{-\gamma}|x|^{-2w}$, the power law of an operator of dimension $w+\gamma$.

Remark 3.20 (Physics notation: BPZ inner product and $Q^\dagger = S$)**.** The radial inner product is the BPZ inner product of radial quantisation, the two-point function of b at the origin with the inversion conjugate of a, i.e. the overlap of the in-state of b with the out-state of a. Its Gram matrix $\mathrm{Gram}_{\mathrm{tree}} + \hbar\,\mathrm{Gram}_1$ is the matrix of two-point coefficients of the operator basis, and Y^{adj} is the Hermitian adjoint for the tree metric. Corollary **??** is the one-loop form of $S = Q^\dagger$, $\bar S = \bar Q^\dagger$ and $K = P^\dagger$, with the signs of Theorem **??**: the one-loop special supercharges are the adjoints of the one-loop supercharges for the one-loop-corrected metric, the commutator with $\mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1$ being the change of the metric. The relation for D_1 is the self-adjointness of the one-loop dilatation operator, hence the reality of the one-loop anomalous dimensions.

Remark 3.21 (Physics notation: higher loops)**.** Order $\hbar^k$ is k loops, order g^{2k} in the normalisation of Remark **??**. At two loops the counterterm vertices of order $\hbar$ inserted into one-loop graphs subtract the one-loop subdivergences, and L_2 contains the two-loop corrections to all superconformal generators, $\frac{1}{2}$ of its component along Δ being the two-loop dilatation operator.